Declaration of content-related cooperation

Between

Alfred Wegener Institute Helmholtz Centre for Polar and Marine Research (AWI) Earth System Complexity Group, Telegrafenberg A45, 14473 Potsdam, Germany represented by **Dr. Alexander Robinson**, Group Leader

and

Universidad Complutense de Madrid (UCM) Departamento de Física de la Tierra y Astrofísica, Facultad de Ciencias Físicas Plaza de Ciencias 1, 28040 Madrid, Spain represented by **Prof. Marisa Montoya**

The Earth System Complexity group at AWI (Dr. Alexander Robinson) and the Paleoclimate and Modeling and Analysis group at UCM (Prof. Marisa Montoya) have a long-standing scientific cooperation in the field of **ice-sheet and climate modelling**, with particular focus on:

- the evolution and long-term stability of the Greenland and Antarctic Ice Sheets,
- coupled ice-sheet / climate / sea-level modelling across glacial-interglacial timescales,
- joint development and application of the Yelmo ice-sheet model, CLIMBER-X and related tools,
- uncertainty quantification and calibration of ice-sheet models against paleo and present-day observations.

The cooperation has resulted in joint peer-reviewed publications, jointly supervised doctoral and master's theses, shared software development, and reciprocal research stays of group members in Potsdam and Madrid.

Both institutions confirm that the guest visits to AWI will serve to intensify and deepen the existing cooperation through:

- on-site collaboration with members of the AWI Earth System Complexity group,
- joint use of AWI's high-performance computing resources for the activities described above,
- knowledge transfer between the two groups (methods, workflows, model code),
- planning of follow-up joint activities (publications, proposals, exchanges, workshops).

4. Signatures

For AWI:

Dr. Alexander Robinson Group Leader, Earth System Complexity Alfred Wegener Institute, Potsdam

Date:

For UCM:

Prof. Marisa Montoya Departamento de Física de la Tierra y Astrofísica Universidad Complutense de Madrid

Date: